

USB3, GigE Vision Development (XML parsing)

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1 USB3 and GigE Vision Description

GigE Vision and USB3 Vision is to specify a Gigabit Ethernet interface and image transmission USB3.0 standard by AIA (Automated Imaging Association) initiated. At present the major manufacturers of automation and machine vision in the implementation of the standards, such as LabView, Halcon, Visionpro etc; if the camera supports the industry standard means to do with these visual software seamless integration.

The Support Vision standard camera has two modes of development, one is based on the manufacturer's SDK development, the other is based Vision interface to develop, since the Vision was designated by the AIA, so the vendors SDK is somewhat different in the name of the development process and interface functions .

2 Interface Description

Listed below are all the interfaces, but these are not all camera models are supported, with the actual subject.

2.1 DeviceControl

UserDefName: camera custom name, length 16 byte string ,

DeviceTemperature : the camera's internal temperature, in degrees Celsius.

2.2 AcquisitionControl

AcquisitionMode: Get image mode, "continuous" is acquired continuously, "singleframe" only obtain one, the equivalent of a soft trigger ;

AcquisitionStart: Camera start getting images ;

AcquisitionStop: The camera stops acquiring images ;

AcquisitionFrameRate: The sampling rate of the camera sensor, only a few stalls adjustment.

AcquisitionFrameRateEnable : When enabled Acquisition Frame Rate Abs value set to take effect, otherwise press to obtain maximum frame rate of the camera.

AcquisitionFrameRateAbs : Set the camera's frame rate, in Hz.

2 .3 TriggerControl

TriggerMode: Set the trigger mode, "on" means open, "off" turn it off. After entering the trigger mode, the image output by the trigger signal to control;

TriggerCount: set the number of times the trigger, when a trigger signal is generated inside the camera will trigger several times sensor, the value is determined by the number of times;

TriggerActivation: Set the trigger input signal is "rising", "falling" "double edge" effective.

TriggerDelay: generating the trigger signal will produce a delay, then the trigger sensor, the unit is us;

ExtTrigJitterTime: External Trig pin jitter elimination, the unit is us;

StrobeMode: Strobe control signal output, "Auto" means the synchronization with the exposure, "Custom" means that you can customize the timing, value is determined by StrobeDelay and StrobeWidth;

StrobeDelay: StrobeMode to "Custom" when entered into force only when the unit is us;

StrobeWidth: StrobeDelay properties and the same;

TriggerSoftware: Software generates a trigger signal.

2 .4 ImageFormatControl

PixelFormat: pixel format, color camera is Bayer, black and white is mono, there 8Bit and 12Bit format, such as BayerRG8, BayerRG12Packed, Mono8;

ReverseX: horizontal flip;

ReverseY: vertical flip;

ResolutionPresetSel: Select the inside the camera preset resolution; When selecting a "Custom", the resolution specified by ResolutionMod, OffsetX, OffsetY, Width, Height these values;

ResolutionMod: "Normal" to full resolution mode, "BINSUM2X" as 2X2 binning, "BINAVER2X" for the 2X2 pixel averaging;

OffsetX: ROI horizontal offset;

OffsetY: ROI vertical offset;

Width: width of the image;

Height: height of the image.

2 .5 ExposureControl

Exposure Auto: automatic exposure, when set to "Continuous" open;

Auto Exposure Target: AE reference value, the larger the value, the higher the brightness of the image;

ExposureTime: set the exposure time, in us; ;

2 .6 AnalogControls

Gain: analog gain in units of times;

Contrast: Contrast, ranging from 0 to 200;

Gamma: Gamma, ranging from 0 to 250 ;

2 .7 ColorControl

ColorTemperatureIndex: Selecting a color temperature;

RGain: Red Gain, 0-4;

GGain: Green gain from 0 to 4;

BGain: Blue gain, from 0 to 4;

WBOnce: A white balance, click once, it will generate a new RGain, GGain, BGain parameters, these three values is the parameter after the white balance correction ;

2 .8 DigitalIOControl

InxMod: input mode x ($x = 0 \sim n$), choose to "GPInput" and "TriggerInput", were common INPUT and trigger functions;

OutxMod: x output mode can be selected as "GPOutput", "StrobeOutput" and "PWMOutput", were common OUTPUT, flash output and PWM output;

GPInx: When the mode corresponding to the pin is set to "GPInput" when can read pin level;

GPOutx: When the mode corresponding to the pin is set to "GPOutput", the value set control pin level;

PWMOutxCycle: set the corresponding pin PWM period, in us, only effective when mode is set to "PWMOutput";

PWMOutxDuty: Set the PWM duty cycle, ranging from 0 to 100.

2 .9 UserSetControl

UserSetSelector: Select the camera's internal parameter set, "Default" for the default parameter set can not be changed, "UserSet1" "UserSet2" these two groups can be set by the user;

UserSetLoad: Loading selected (UserSetSelector) set of parameters;

UserSetSave: save the current settings to selected (UserSetSelector) set of parameters;

UserSetDefaultSelector: when the camera is powered on, the default parameter set is loaded.